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COMPUTER

CHAPTER 26

Microsoft Excel

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COMPUTER · CHAPTER 26

Microsoft Excel

ਮਾਈਕ੍ਰੋਸਾਫਟ ਐਕਸਲ

IN THIS CHAPTER

- Spreadsheet Structure
- Workbook, Worksheet And Cell
- Relationship
- Structure Of Excel Window
- Important Parts Of Excel
- Ribbon Tabs In Excel

Microsoft Excel (MS Excel) is a spreadsheet application developed by Microsoft. It is used to store, organize, calculate, analyze, and visualize data in the form of rows and columns.

MS Excel is one of the most widely used **data processing and data analysis tools** in offices, businesses, schools, banks, and government organizations.

MS Excel is a spreadsheet software used to perform calculations, create tables, analyze data, generate charts, and prepare reports.

Nature of MS Excel = Spreadsheet Application / Data Processing Software

WHAT IS A SPREADSHEET?

A spreadsheet is a table made up of **rows and columns**.

The intersection of a row and a column forms a **cell**.

SPREADSHEET STRUCTURE

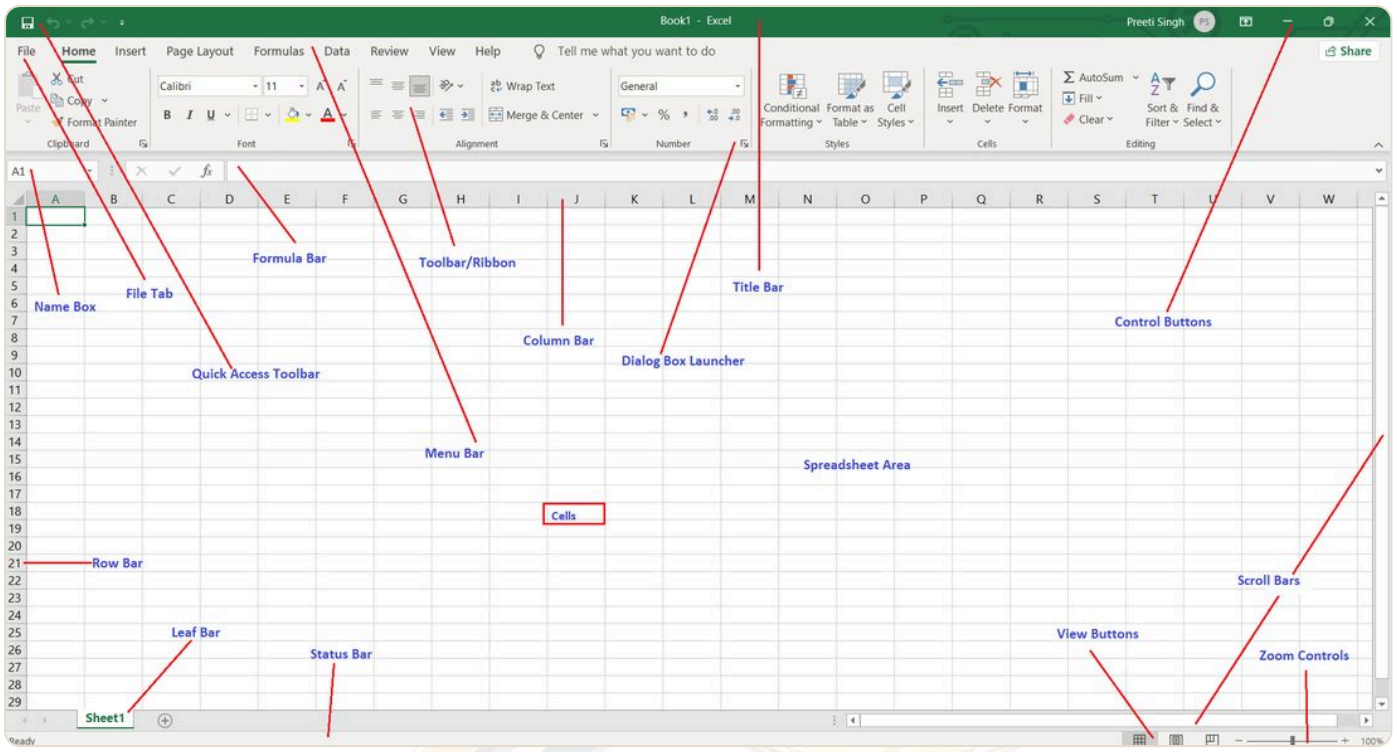
Cell = Intersection of Row and Column

Example:

Cell Address = B3

(B = Column, 3 = Row)

WORKBOOK, WORKSHEET AND CELL



Workbook

An Excel file is called a **Workbook**.

Example

Book1.xlsx

Worksheet

A single sheet inside a workbook is called a Worksheet.

Example

Sheet1, Sheet2, Sheet3

Cell

A rectangular box formed by the intersection of a row and a column.

Example

A1, B5, D10

RELATIONSHIP

Workbook

│

├── Worksheet 1

├── Worksheet 2

└── Worksheet 3

│

└── Cells

Exam Point

Collection of Cells → Worksheet

Collection of Worksheets → Workbook

STRUCTURE OF EXCEL WINDOW

The Excel window consists of the Title Bar, Quick Access Toolbar, Ribbon, Name Box, Formula Bar, Column and Row Headings, Spreadsheet Area, Sheet Tabs, Status Bar and Zoom Control.

IMPORTANT PARTS OF EXCEL

1. Name Box

Displays the address of the currently selected cell.

Example

If cell D5 is selected:

Name Box = D5

2. Formula Bar

Used to enter and edit formulas and functions.

Example

=SUM(A1:A10)

3. Column Heading

Columns are identified by letters.

ABCDEF...

Last Column

XFD

Total Columns

16,384

Exam Point

Excel Columns = A to XFD

4. Row Heading

Rows are identified by numbers.

1

2

3

4

Last Row

1,048,576

Exam Point

Total Rows = 1,048,576

5. Sheet Tabs

Displays all worksheets in the workbook.

Example:

- Sheet1
- Sheet2
- Sheet3

6. Spreadsheet Area

The working area where data is entered.

7. Zoom Control

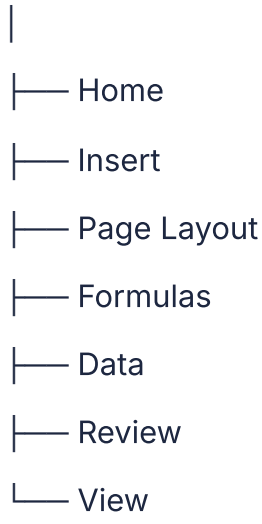
Used to zoom in and zoom out.

Range

10% to 400%

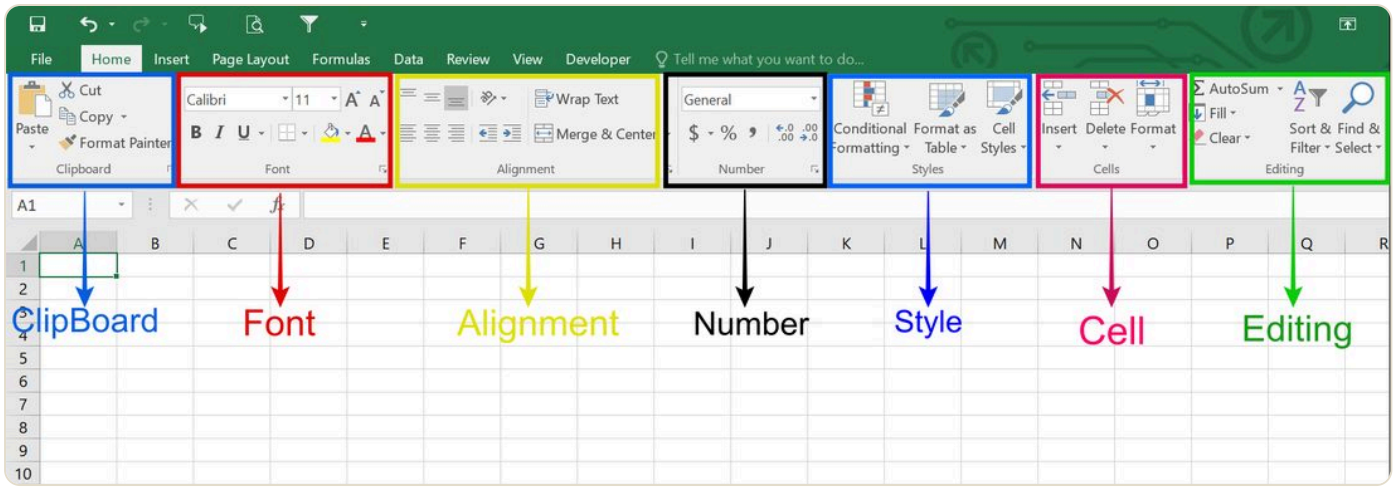
RIBBON TABS IN EXCEL

MS EXCEL



HOME TAB

The Home Tab contains the most frequently used commands.



Groups in Home Tab

Home

|

├── Clipboard

├── Font

├── Alignment

├── Number

├── Styles

├── Cells

└── Editing

1. CLIPBOARD GROUP

Contains:

- Cut
- Copy
- Paste
- Format Painter

Format Painter

Copies formatting from one cell to another.

2. FONT GROUP

Used for formatting text.

Options

- Font Style
- Font Size
- Bold
- Italic
- Underline
- Font Color
- Fill Color

3. ALIGNMENT GROUP

Controls alignment of cell contents.

Features

- Left Align
- Center Align
- Right Align
- Top Align
- Bottom Align

Wrap Text

Displays long text on multiple lines within a cell.

Example:

Gur preet

Singh

Merge & Center

Combines multiple cells into one cell and centers the data.

Example:

A1+B1+C1

↓

Merged Cell

4. NUMBER GROUP

Used to format numbers.

Examples

- Currency (₹)
- Dollar ($)
- Percentage (%)
- Date
- Time

5. STYLES GROUP

Contains:

Conditional Formatting :Formats cells according to conditions.

Example:

Marks > 90 → Green

Marks < 40 → Red

Cell Styles :Predefined formatting styles.

Format as Table :Converts selected data into a table.

6. CELLS GROUP

Used for:

- Insert Cells
- Delete Cells
- Format Cells

7. EDITING GROUP

Contains:

Sort :Arranges data.

Example:

A to Z

Z to A

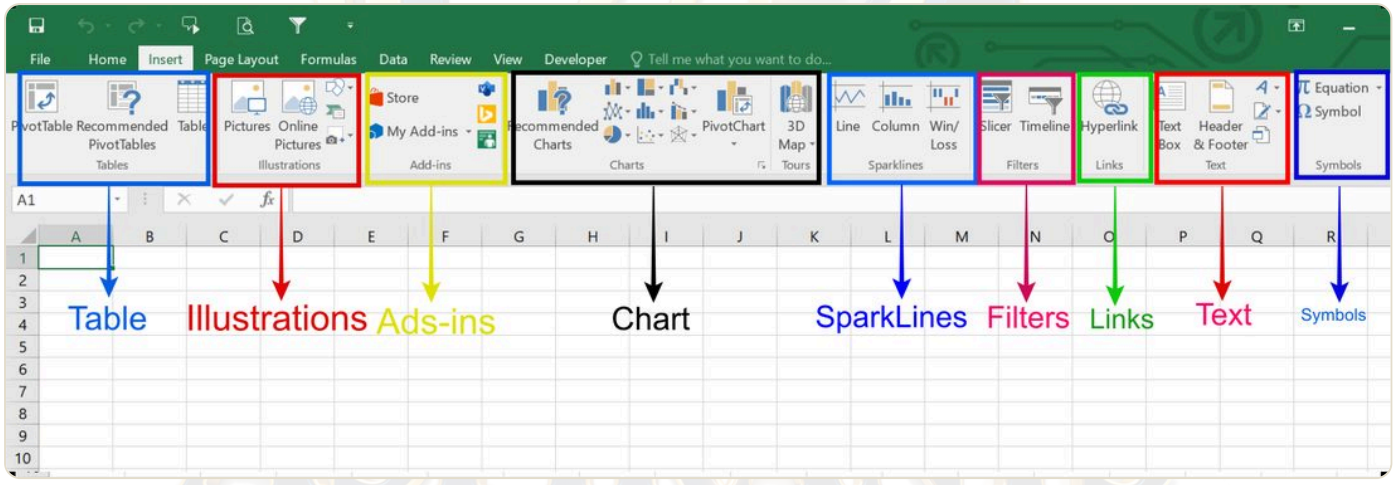
Filter :Displays only selected data.

Find & Replace :Search and replace data.

Clear :Removes contents or formatting.

INSERT TAB

Used to insert various objects.



1. Tables

Creates tables.

Pivot Table :A special table used to summarize, analyze, and organize large amounts of data.

Example

Sales data of 1000 employees summarized into one report.

Exam Point

Pivot Table = Summary of Complex Data

2. Filters

Slicer :Visual filtering tool for Pivot Tables.

Timeline :Filters Pivot Table data based on dates.

3. Charts

Creates:

- Bar Charts
- Pie Charts
- Line Charts
- Area Charts

4. Text Group

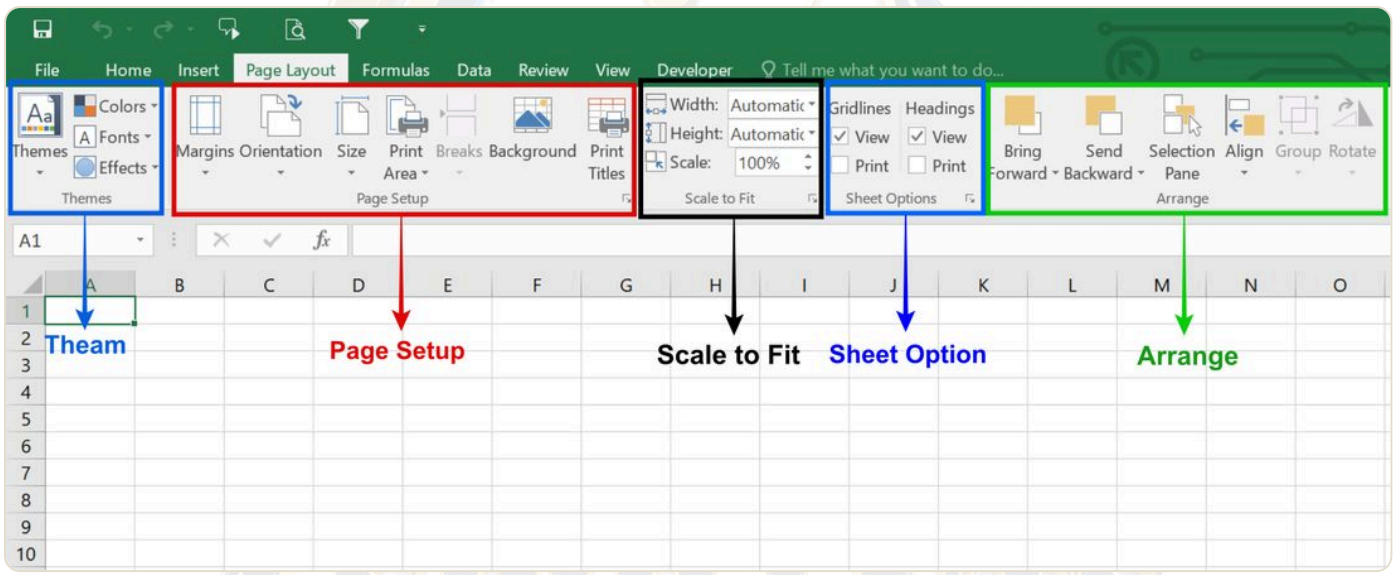
Contains:

- Header
- Footer
- WordArt
- Signature Line

NoteMost of the features in the Insert tab are similar to the Insert tab in MS Word, so you can learn them from there.

PAGE LAYOUT TAB

Controls page appearance.



The **Page Layout** tab in Microsoft Excel allows users to control how their workbook will appear when printed or viewed in different layouts. The key sections are:

1. Themes

• Changes the overall appearance of the workbook.

• Includes fonts, colours, and effects that give a consistent look across all sheets.

2. Page Setup

• **Margins**: Adjusts the white space around the page.

• **Orientation**: Choose between **Portrait** (default) or **Landscape**.

• **Page Breaks**: Splits the worksheet into multiple printable pages.

3. Sheet Options

- Gridlines and Headings (View / Print):

- Gridlines: Toggle the display of worksheet gridlines.
- Headings: Show or hide row and column headings.

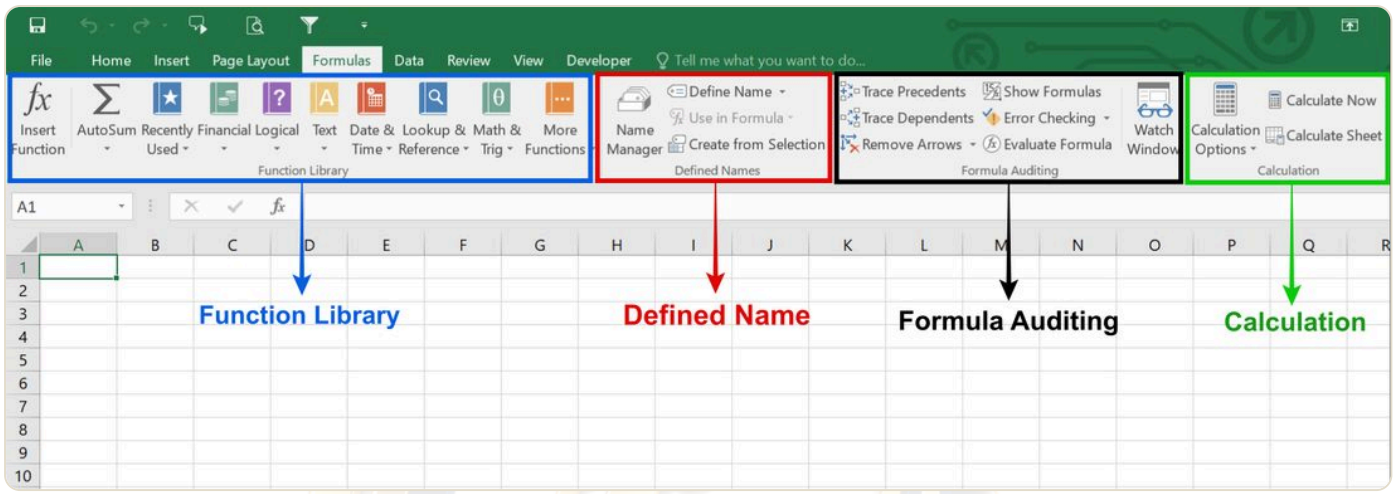
4. Arrange

• Used to arrange workbook windows for easier viewing and management.

These settings help in controlling the printing and presentation of data effectively.

FORMULAS TAB

Contains all formulas and functions.



FUNCTION

A predefined formula in Excel is called a Function.

Example

=SUM(A1:A10)

Exam Point

Function = Predefined Formula

Common Functions

1. SUM

Adds numbers.

=SUM(A1:A10)

2. AVERAGE

Finds average.

=AVERAGE(A1:A10)

3. MAX

Largest value.

=MAX(A1:A10)

4. MIN

Smallest value.

=MIN(A1:A10)

5. COUNT

Counts numerical entries.

=COUNT(A1:A10)

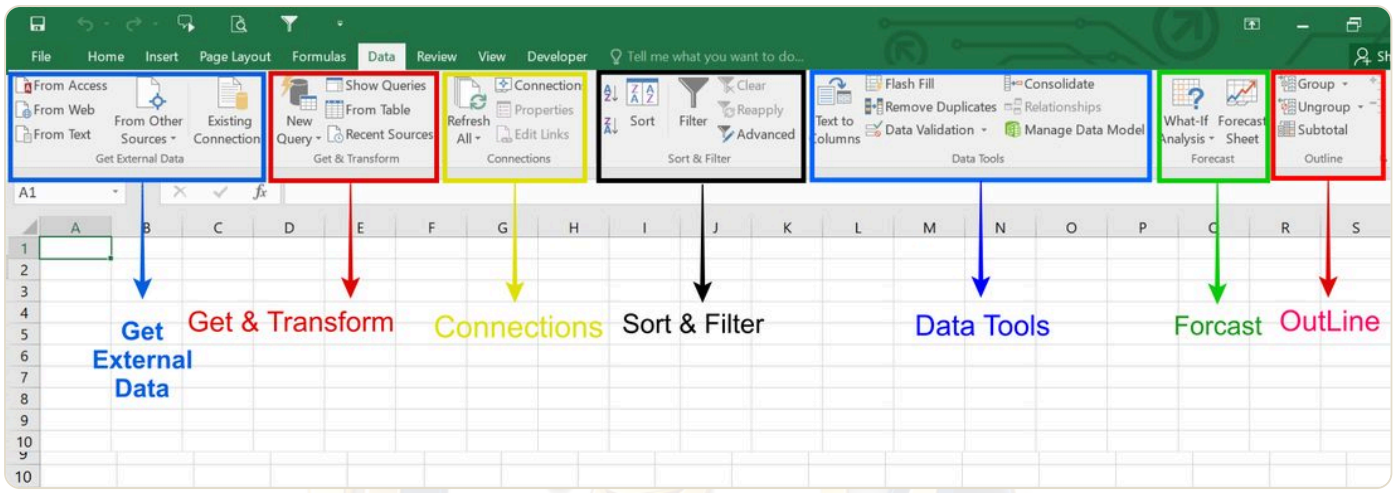
Exam Point :Every Formula Starts With "="

Example:

=A1+B1

DATA TAB

Used for data management.



Features

- Sort
- Filter
- Data Validation
- Remove Duplicates
- External Data Import

The Data tab in Microsoft Excel is one of the most important sections for managing, analysing, and organising data efficiently. It provides a wide range of tools to import, sort, filter, and process data from various sources. Here is an overview of the key groups and commands in the Data tab:

1. Get & Transform Data (Get External Data)

- Get Data: Allows you to import data from external sources like databases, web pages, text/CSV files, XML, JSON, or other workbooks.
- Connections: Manage the links between your Excel file and connected data sources.
- Refresh All: Update data from connected sources.

2. Sort & Filter

- Sort Ascending / Descending: Arrange data either from smallest to largest or largest to smallest.
- Custom Sort: Sort data by multiple columns or custom order.
- Filter: Add filters to columns to view only the data you need.
- Clear: Remove applied filters.
- Reapply: Reapply the last used filter settings.

3. Data Tools

- Text to Columns: Split data in a single column into multiple columns based on delimiters (like commas or spaces).
- Remove Duplicates: Quickly remove repeated values.
- Data Validation: Restrict the type of data that can be entered in a cell (like dates, numbers, or specific lists).
- Consolidate: Combine data from multiple ranges.
- What-If Analysis: Includes tools like Scenario Manager and Goal Seek to analyse different outcomes.

4. Forecast (Data Analysis)

- Forecast Sheet: Create projections and future trends based on historical data.

5. Outline

- Group / Ungroup: Arrange data in collapsible sections.
- Subtotal: Automatically calculate sums, averages, or other aggregations for grouped data.

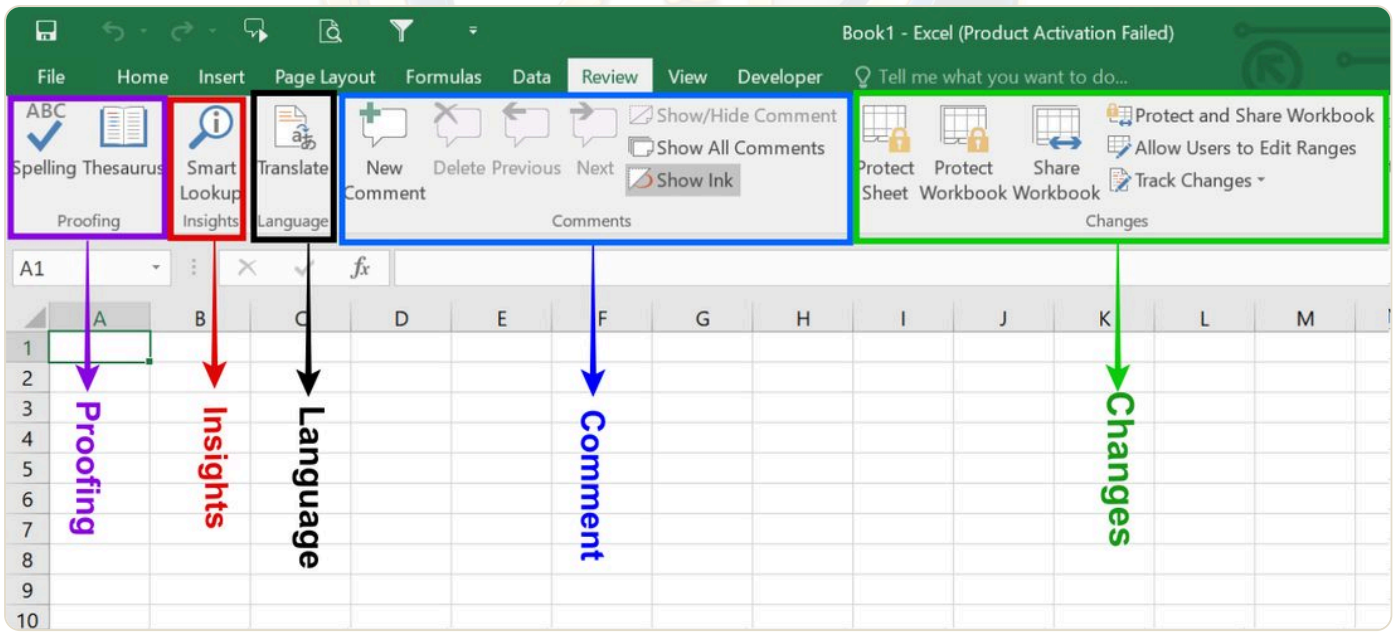
6. Queries & Connections (Power Query)

- Lets you manage all data queries in one place, especially for large or complex datasets.

The Data tab is essential when working with large datasets, performing analysis, and keeping your Excel workbook organised and dynamic.

REVIEW TAB

Similar to MS Word Review Tab.



Features

- Spelling Check
- Comments
- Track Changes
- Protect Sheet
- Protect Workbook

1. Proofing

- Spelling: Checks the worksheet for spelling errors.

- Thesaurus: Offers synonyms for selected words.
- Research/Smart Lookup: Provides definitions, context, and related information for the selected text.

2. Accessibility

- Check Accessibility: Scans the workbook for elements that may be difficult for people with disabilities to read or use.

3. Comments

- New Comment: Inserts a comment in a cell to provide notes or instructions.
- Show/Hide Comment: Displays or hides comments in selected cells.
- Show All Comments: Displays all comments in the worksheet.
- Delete Comment: Removes a comment from a cell.
- Previous/Next Comment: Navigates through the comments in the worksheet.

4. Notes (In newer versions of Excel)

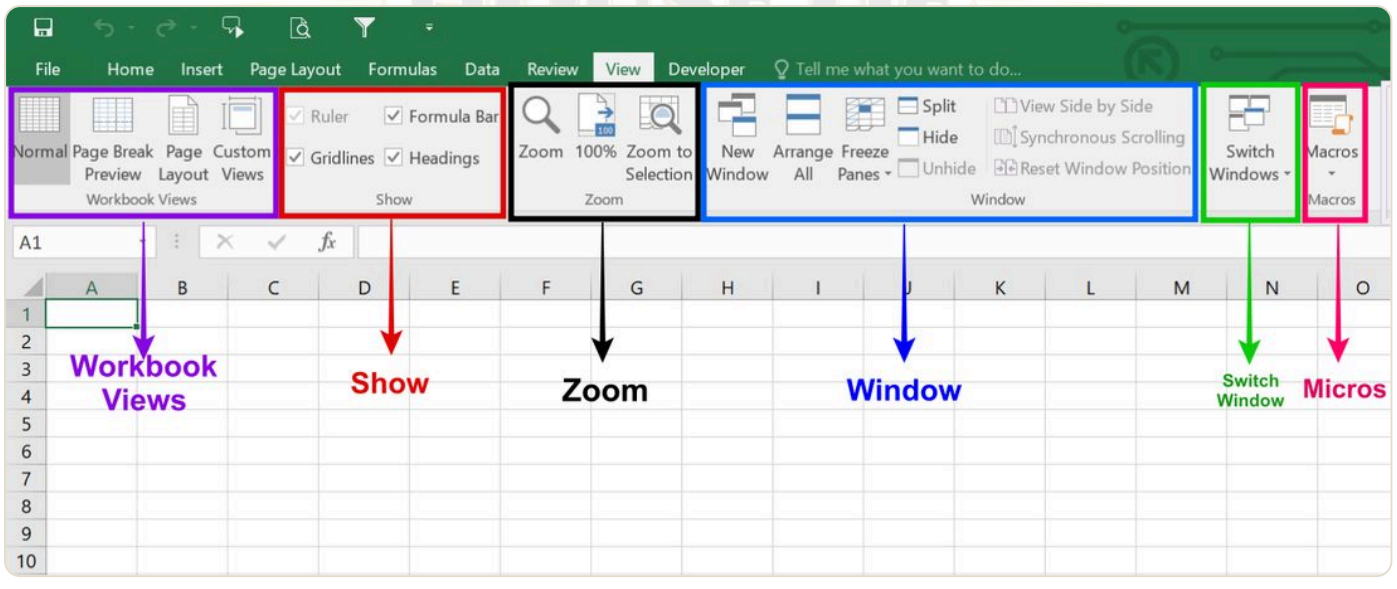
- New Note: Adds a traditional note to a cell.
- Edit Note: Modifies existing notes.
- Convert Notes: Converts old-style notes to modern comments.

5. Protect

- Protect Sheet: Locks the worksheet to prevent users from changing data or structure.
- Protect Workbook: Secures the workbook structure to prevent adding, deleting, or moving sheets.
- Allow Edit Ranges: Specifies which cells can be edited while the sheet is protected.

The Review tab is mainly for data verification, collaboration, and security. For team workbooks, this tab helps monitor changes, leave feedback, and ensure the document is error free before sharing.

VIEW TAB



Workbook Views

- Normal View: Default View
- Page Break Preview :Shows page breaks.
- Page Layout View :Shows worksheet as printed pages.
- Custom View:Creates customized worksheet views.

Show Group

Show/Hide:

- Formula Bar
- Gridlines
- Headings

Window Group

Freeze Panes :Locks rows and columns while scrolling.

Example

Keep first row visible while scrolling.

Exam Point

Freeze Panes = Lock Rows and Columns

CELL ADDRESS

A cell address is identified by: Column + Row

Example

D5

D = Column

5 = Row

RANGE OF CELLS

A group of cells is called a Range.

Example

A1:C3

Meaning :Includes all cells from A1 to C3.

Exam Point :Range is represented using Colon (:)

DEFAULT SETTINGS

Feature	Value
Default Workbook Name	Book1
Default Orientation	Portrait
Default View	Normal
Default File Extension	.xlsx
Mouse Pointer Shape	Plus (+)

DATA ALIGNMENT

Data Type	Default Alignment
Text	Left
Numbers	Right

Exam Point

Text → Left

Numbers → Right

EXCEL FILE EXTENSIONS

Extension	Purpose
.xls	Older Excel Workbook
.xlsx	Default Workbook
.xlsm	Macro-enabled Workbook
.xla	Add-in
.xlam	Macro-enabled Add-in
.xlw	Workspace File
.csv	Comma Separated Values
.ods	OpenDocument Spreadsheet
.txt	Text File
.html	Web Page

COMMON EXCEL ERRORS

Error	Meaning
#DIV/0!	Division by Zero
#NAME?	Unknown Function Name
#VALUE!	Wrong Data Type
#REF!	Invalid Cell Reference
#NUM!	Invalid Number
#N/A	Value Not Available
#####	Column Too Narrow

Exam Point

Most Common Error = #DIV/0!

WORKBOOK VS WORKSHEET

Workbook	Worksheet
Excel File	Sheet Inside Workbook
Example: Book1.xlsx	Example: Sheet1

SORT VS FILTER

Sort	Filter
Rearranges Data	Hides Unwanted Data
A→Z	Show Specific Records

Shortcut keys	Description
ctrl+right	To go to last column
Ctrl +left	To go to first column
Ctrl +up	To go to first row
Ctrl +down	To go to last row
shift+arrows	To select cells
Ctrl + N	Create new workbook
Shift + F11	Create new sheet
Ctrl +1	Format cell dialog box
Ctrl +shift + +	To add new column and row
Ctrl + -	To delete selected row and column
alt+page up/down	Horizontal scrolling
Page up/down	Vertical scrolling
ctrl+page up/down	To shifts between sheets
Ctrl +shift +page up/down	Select all sheets
Ctrl +space	To select entire column
Shift + space	To select entire row
Ctrl +shift+space	To select all sheet

Shortcut keys	Description
Ctrl +home	To move to first cell
Ctrl +end	To move to last cell
Ctrl +shift +f	Format cell dialog box
Ctrl +alt+end	Select all used cells in a sheet
Ctrl + G	Open Goto dialog box
Ctrl +F11	To add macro sheet
Alt + enter	To write in next line in same cell
Enter	Move cursor to next cell
F11	To add chart in different sheet
Ctrl +shift +U	To expand formula bar
Ctrl + tab ,ctrl +f6	To switch between workbooks
Ctrl +shift +$	To add currency sign
ctrl +shift+#	To show date format
Ctrl +shift +@	To show time format
ctrl + ;	Current date
Ctrl +shift+:	Current time
Ctrl + t	To create table
Ctrl +enter	To move the text in next line in a cell
F1	HELP
Ctrl + f1	To hide ribbon
ctrl +shift + f1	Hide all bar
F2	To edit active cell
Shift + f2	To add comment in a cell
Ctrl +shift +%	To add % sign
Ctrl + 1	Cell dialog box
Shift + f3	Insert function dialog box
Shift +f5	To open find and replace box
ctrl +5	Strike through
F7	Spelling and grammar
ctrl + 9	Hide row
Ctrl +shift+9	Show row

Shortcut keys	Description
F11	To insert chart
Alt +F11	To open Visual Basic Editor (Alt + F1 adds chart in same sheet)
Shift +F11	To add new sheet

Formulas	Description
Sum(A1:E1)	It add all the number which include A1,b1,c1,d1,e1
sum(A1,E1)	It add the numbers which include A1 ,E1 only
Min()	To find minimum number
Max()	Maximum number
average()	Average of numbers
if(condition,print1,print2)	If condition satisfied print1 otherwise print 2
Rank()	To find rank from selected cells
concatenate(a2,b2)	To join value of two cells in a single cell
now()	Today date and time
today()	Present date only
count()	No of cells which have numerical value
countA()	It count no of cells which have numerical, alphabet value
countblank()	Count number of blank cells
frequency(a1:c1,binarray)	To count the number of duplicate value
lookup()	Used for look ups. Value from a one row and one column
VLookup()	Look up a value in the leftmost column of a table
Hlookup()	Look up a value in the top row of a table
Sum product	First multiply then add
clean()	It avoids unprintable character
trim()	Remove unwanted spaces in text
lower()	Convert the character of text in lower case
Upper()	Convert to upper case
proper()	Convert into proper case
len()	It return the length of a text
find()	Find a character in the text
REPT()	To repeat a text a given number of times

Formulas	Description
substitute(), replace()	Replace a selected character with another character
exact()	To check whether two text are similar or not
T()	To check the text is there in a cell or not
Sqrt()	Square root
large()	Return the largest value from various numbers
small()	Return the small value
round()	Round of the numbers who are in points

MOST IMPORTANT EXAM FACTS

1. MS Excel is a Spreadsheet Application.
2. Default workbook name = Book1.
3. Default extension = .xlsx.
4. Total columns = 16,384 (A–XFD).
5. Total rows = 1,048,576.
6. Cell = Intersection of Row and Column.
7. Formula starts with "=".
8. Function = Predefined Formula.
9. SUM is the most commonly used function.
10. Pivot Table summarizes large data.
11. Freeze Panes locks rows and columns.
12. Merge & Center combines cells.
13. Conditional Formatting formats data automatically.
14. Text aligns left by default.
15. Numbers align right by default.

QUICK REVISION NOTES

- MS Excel = Spreadsheet Software
- Workbook = Excel File
- Worksheet = Sheet
- Cell = Row + Column Intersection
- Default File = Book1
- Default Extension = .xlsx
- Columns = A to XFD
- Rows = 1,048,576

- Formula starts with "="
- SUM = Addition Function
- Pivot Table = Data Summary
- Sort = Arrange Data
- Filter = Display Specific Data
- Conditional Formatting = Format Based on Conditions
- Freeze Panes = Lock Rows/Columns
- Merge & Center = Combine Cells
- Text Left, Numbers Right